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Cambro-Ordovician paleogeography of the Southeastern New England Avalon Zone: Implications for Gondwana breakup

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ABSTRACT

Paleomagnetic measurements and U-Pb geochronology of alkalic igneous rocks intruding the Ediacaran Cambridge Argillite and the Lower Cambrian Weymouth Formation north of Boston at Nahant, Massachusetts yield a well-constrained paleopole. The oldest intrusive phases are a syenodiorite sill and a granite dike with $^{206}Pb/^{238}U$ crystallization ages of 490.19 ± 0.90 Ma and 489.03 ± 0.82 Ma, respectively. This sequence along with undated mafic sills penetrating both the argillite and Weymouth strata are cut by 488.53 ± 0.81 Ma gabbro locally associated with 488.48 ± 0.79 Ma syenite forming a steep-sided intrusive plug. Ten paleomagnetic sampling sites within the mafic sills and the gabbro contain a stable A magnetic component with a mean direction of D = 278.5°, I = –76.7° (A_{95} = 3.9°, N = 10). This is considered to be the primary magnetization direction because it passes both fold and reversals tests. The corresponding paleopole at 320°E, 34°N (A_{95} = 7.2°, N = 10) places the Southeastern New England Avalon Zone at a southerly paleolatitude of 65° slightly removed from the African margin of Gondwana and suggests that bimodal Nahant magmatism is linked with latest Cambrian–Early Ordovician rifting of West Avalonia to form the Rheic Ocean.

Stable B and C magnetic components in Nahant syenodiorite and syenite give rise to virtual geomagnetic poles that track Late Silurian through Carboniferous segments of the North American apparent polar wander path, indicating docking of West Avalonia by mid-Paleozoic time.

INTRODUCTION

Avalonia is a collection of terranes that originated along northern Gondwana as belts in which late Neoproterozoic arc magmatism was succeeded by Cambrian shelf sedimentation (recent overviews in Murphy et al., 2004; Hibbard et al., 2006). Among the terranes sharing this history are portions of Great Britain and Europe (East Avalonia of Soper and Woodcock, 1990), as well as Atlantic portions of Newfoundland, New Brunswick, Nova Scotia, and southeastern New England (West Avalonia beginning with Nance and Murphy, 1994; inset in Fig. 1). These and other peri-Gondwanan terranes (Ganderia of Hibbard et al., 2006) began to separate from the Gondwanan margin by Late Cambrian–Early Ordovician time, so that West Avalonia came to lie simultaneously south of Iapetus and north of the Rheic Ocean, which was opening to the south (Van der Voo, 1988; Scotese and McKerrow, 1991; van Staal et al., 1998; Cocks and Torsvik, 2002; Murphy et al., 2006). Cold water faunas (Scotese and McKerrow, 1990; Fortey and Cocks, 1992; Fortey and Cocks, 2003) and paleopoles from Gondwana (Grunow and Encarnacion, 2000; Torsvik and Van der Voo, 2002; older poles reviewed in Van der Voo, 1988) establish the high latitude setting for these events, and paleomagnetic data from northern Appalachian terranes that originated as volcanic arcs within Iapetus (Mac Niocaill et al., 1997 and references therein) place additional constraints on the position of West Avalonia with respect to this framework. In particular, West Avalonia must lie farther south than the most southerly of the arc-related sequences represented by Middle Ordovician rocks of the Miramichi terrane in northern New Brunswick that formed at a paleolatitude of 53°S (Liss et al., 1993; peri-Avalonian arc of Mac Niocaill et al., 1997; peri-Gondwanan arc system located outboard of Gander margin in Hibbard et al., 2006). Direct paleomagnetic constraints positioning Avalonia at this time have come largely from British localities in East Avalonia (paleopoles in Table 2 in Van der Voo, 1988 and Table 1 in Cocks and Torsvik, 2002). Indeed the only West Avalonian paleopole used in these reconstructions is based on one of five disparate, structurally uncorrected directions from the Nahant Gabbro intruding Ediacaran argillites and fossiliferous Lower Cambrian strata overlying the Avalonian arc sequence in southeastern New England (Weisse et al., 1985; Fig. 1). The study presented below reexamines the gabbro and related intrusive rocks at Nahant to obtain a precisely dated, structurally corrected Cambro-Ordovician paleopole that fixes the peri-Gondwanan position of West Avalonia at the onset of Rheic Ocean opening.

GEOLOGY OF NAHANT

Igneous rocks at Nahant, Massachusetts, are shown as undifferentiated Nahant Gabbro (named by Lane, 1888) on previously published geologic maps (Emerson, 1917; LaForge, 1932; Zen, 1983), but other rock types make up a compositionally diverse plutonic suite. Major-element geochemistry (Washington, 1899; Perry, 1923; Acaster and Bickford, 1999; Ross and Bailey, 2001) including new data from this study (GSA Data Repository Table DR1[1]) indicate alkalic to subalkalic compositions ranging from 44% to 72% silica by weight with a gap between 50% and 60% (solid circles in Fig. 2A). The principal siliceous members of the plutonic suite are syenodiorite at Little Nahant and syenite at Saunders Ledge on the north side of Nahant proper (Fig. 1; plutonic nomenclature in Fig. 2A by Wilson, 1989). Peridotites and pyroxenites are also reported locally within the central body of gabbro delineated by the dashed contact in Figure 1 (Clapp, 1921; Kaye, 1965; Verma, 1973; Ross and Bailey, 2001).

The Weymouth Formation rims the eastern side of the Nahant Gabbro at East Point (Fig. 1) where metasiltstones are interbedded with limestones containing small shelly fossils that

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[1]GSA Data Repository item 2009186, major and trace element geochemistry of intrusive rocks and pelitic hornfels at Nahant, Massachusetts, is available at http://www.geosociety.org/pubs/ft2009.htm or by request to editing@geosociety.org.

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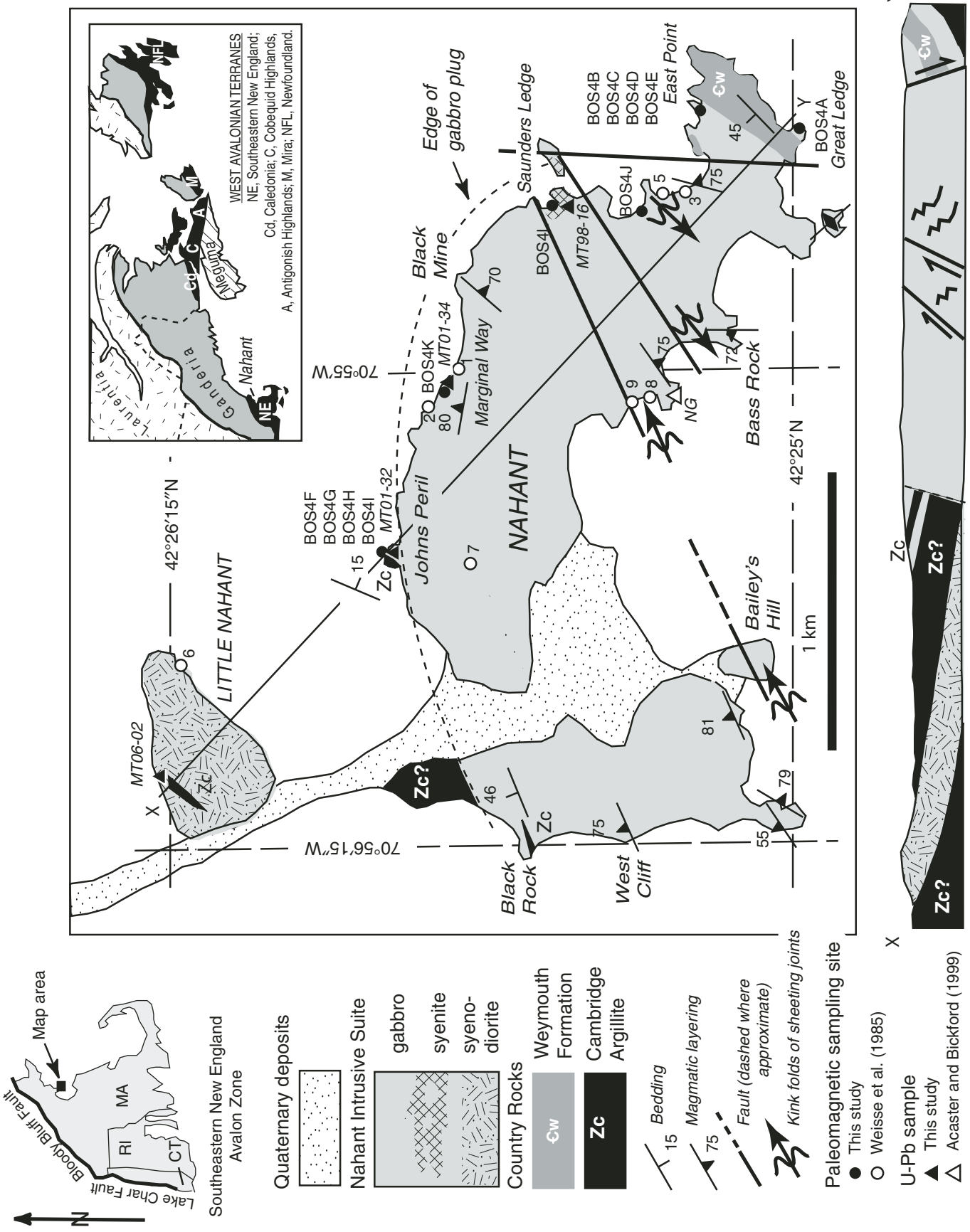


Figure 1. Geologic sketch map and cross section of Nahant, Massachusetts. Inset showing distribution of West Avalonian terranes modified from Potter et al. (2008).

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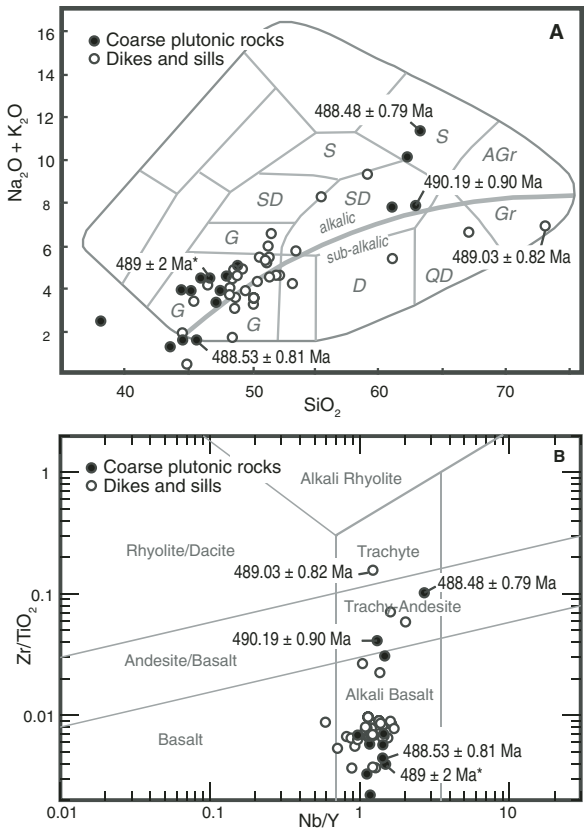


Figure 2. Major- and trace-element geochemistry of intrusive igneous rocks at Nahant, Massachusetts. Plots include published analyses (Washington, 1899; Perry, 1923; Acaster and Bickford, 1999; Ross and Bailey, 2001) and analyses from this study (Table DR1 [see footnote 1]). Dates are shown for samples dated by U-Pb geochronology (starred value is upper concordia intercept date from Acaster and Bickford, 1999; others are $^{206}Pb/^{238}U$ dates from this study). (A) Total alkali versus silica diagram (Cox et al., 1979) with plutonic nomenclature after Wilson (1989) and dividing line between alkalic and subalkalic magma series after Miyashiro (1978). (B) Modified Winchester diagram (Winchester and Floyd, 1977; Pearce, 1996).

have long demonstrated a Lower Cambrian age (Foerste, 1889; Grabau, 1900; Landing, 1996). To the north at Johns Peril (Fig. 1), the contact rock is unfossiliferous black hornfels (lydite of Lane, 1890) for which a Cambrian age can only be assumed (LaForge, 1932; Zen, 1983). The same is true of unfossiliferous green siltstone and quartzite engulfed by syenodiorite on the north side of Little Nahant (Lane, 1890; Verma, 1973). Geochemical distinctions between these units and Weymouth strata (Table DR2 [see footnote 1]) suggest that they are better interpreted as the top of the Cambridge Argillite that underlies the north half of the Boston Basin and most of Boston Harbor (570 Ma maximum age and argillite geochemistry in Thompson and Bowring, 2000).

Older plutonic units as well as the metasedimentary rocks described above are cut by numerous dikes and sills (open circles in Figs. 2A and 2B) with thicknesses (tens of cm to less than 10 m) too small to show at the scale of Figure 1. These are most fully documented on the eastern end of Nahant between East Point and Great Ledge where they have been subdivided into alkalic and tholeiitic types (Ross and Bailey, 2001). A common origin seems likely, however, because all of them plot with coarse-grained members of the plutonic suite in the alkalic field of the modified Winchester and Floyd immobile trace-element diagram (Fig. 2B). Following Clapp (1921), the syenodiorite of Little Nahant is shown as a sill (Fig. 1) with a thickness here estimated at ~130 m and a dip parallel to strata at Johns Peril.

The dominant structure at Nahant has generally been described as a syncline because metasedimentary units bordering the gabbro dip moderately northwest between East Point and Great Ledge (Fig. 1) and moderately southeast at Johns Peril. Previous workers have interpreted the main gabbro as a much thicker sill that was intruded either during or after folding (Lane, 1890; Clapp, 1921; Weisse et al., 1985), but steep magmatic layering throughout the gabbro of central Nahant (inside dashed line in Fig. 1) indicates a discordant, plug-like relationship with the tilted wall rock (cross section in Fig. 1). Assuming that layering in the wall rock was originally horizontal, this relationship implies that tilting preceded gabbro intrusion.

Ubiquitous gently inclined sheeting joints that developed during unroofing of the Nahant plutonic suite are locally contorted into asymmetric, south-verging kink folds (described in detail by Kaye, 1965) in zones where north-dipping faults are also observed. These folds are interpreted here as the result of motions on reverse faults post-dating the formation of the joints.

The only available U-Pb age constraint for the intrusive suite described above comes from gabbro on the south side of Nahant (Acaster and Bickford, 1999; NG in Fig. 1). A U-Pb upper intercept date of 489 ± 2 Ma was obtained from four discordant multigrain zircon fractions analyzed by isotope dilution and thermal ionization mass spectrometry (ID-TIMS); no zircon pretreatment was reported.

AGE OF NAHANT INTRUSIVE SUITE

U-Pb single-zircon dates in this study come from gabbro, syenodiorite, syenite, and granite representing the full compositional spectrum of coarse-grained intrusive rocks at Nahant. Sample settings and rock descriptions are presented below, and geographic coordinates of sample locations are listed in Table 1. Zircons separated from the four crushed samples by standard Wilfley table, magnetic and heavy liquids techniques were handpicked under a binocular microscope and then pretreated by the thermal annealing and chemical leaching method (chemical abrasion thermal ionization mass spectrometry [CA-TIMS], Mattinson, 2005) that effectively eliminates Pb loss. Pretreated grains were spiked with a mixed ^{205}Pb-^{233}U-^{235}U tracer solution and dissolved in HF. Dissolved Pb and U were separated using an HCl-based ion exchange chemistry procedure (modified after Krogh, 1973), loaded onto single, degassed Re filaments together with a silica gel–H_3PO_4 emitter, and their isotopic compositions were measured on the VG Sector 54 multicollector thermal ionization mass spectrometer at the Massachusetts Institute of Technology. Isotopic data from the Nahant samples along with details of fractionation and blank corrections appear in Table 1. The U-Pb data reduction and calculation of ages and their internal uncertainties follow in general the error propagation algorithms of Ludwig (1980) and the program Isoplot (Ludwig, 2005) using the U decay constants of Jaffey et al. (1971). Calculated U-Pb dates are reported at 95% confidence level as summarized in Table A1 and plotted on standard concordia plots of Figures 3A–3D.

With the exception of two outlier analyses, the CA-TIMS zircon analyses of this study produced coherent clusters of data for which all scatter lies within the two-sigma analytical uncertainty (analogous to *equivalent* data of Ludwig, 1998). Since all coherent clusters of high-precision zircon data exhibit a slight but systematic age discordance (manifested by older $^{207}Pb/^{206}Pb$ dates relative to the corresponding Pb/U dates) possibly due to inaccuracy in one or both of the U decay constants (Mattinson, 2000; Schoene et al., 2006), we consider the weighted mean

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TABLE 1. U-PB ISOTOPIC DATA FOR ALKALIC IGNEOUS ROCKS, NAHANT, MASSACHUSETTS

Sample fractions[†]	Pb_c[§] (pg)	Pb*[§] Pb_c	Th U	^{206}Pb[#] ^{204}Pb	^{208}Pb[††] ^{206}Pb	Ratios						Age (Ma)			Corr. coef.
						^{206}Pb[§§] ^{238}U	Error (2σ%)	^{207}Pb[§§] ^{235}U	Error (2σ%)	^{207}Pb[§§] ^{206}Pb	Error (2σ%)	^{206}Pb ^{238}U	^{207}Pb ^{235}U	^{207}Pb ^{206}Pb	
Syenite, Saunders Ledge (MT98-16 new) (GPS coordinates: N42° 25.498′, W070° 54.572′)															
z1	1.6	45.3	0.75	2574.6	0.236	0.078742	(0.09)	0.61874	(0.17)	0.05699	(0.14)	**488.61**	**489.06**	**491.2**	0.53
z2	0.7	138.2	0.67	7967.1	0.212	0.078727	(0.08)	0.61858	(0.13)	0.05699	(0.10)	**488.52**	**488.95**	**491.0**	0.65
z3	0.9	144.3	0.67	8335.0	0.209	0.078716	(0.08)	0.61775	(0.13)	0.05692	(0.09)	**488.46**	**488.43**	**488.3**	0.66
z4	0.9	257.0	0.91	13969.5	0.286	0.078702	(0.08)	0.61812	(0.12)	0.05696	(0.09)	**488.37**	**488.67**	**490.1**	0.68
z5	0.9	128.8	0.77	7255.5	0.241	0.078714	(0.08)	0.61811	(0.13)	0.05695	(0.10)	**488.44**	**488.66**	**489.7**	0.65
Felsic dike, Johns Peril (MT01-32 new) (GPS coordinates: N42° 25.785′, W070° 55.520′)															
z1	0.5	27.0	0.64	1584.9	0.202	0.078804	(0.09)	0.61959	(0.24)	0.05702	(0.21)	**488.98**	**489.59**	**492.4**	0.44
z2	0.5	20.7	0.76	1180.2	0.241	0.078782	(0.10)	0.61954	(0.32)	0.05703	(0.29)	**488.85**	**489.56**	**492.9**	0.41
z3	1.4	11.4	0.75	659.4	0.241	0.078835	(0.10)	0.62200	(0.47)	0.05722	(0.43)	**489.16**	**491.10**	**500.1**	0.40
z6	0.5	22.4	0.49	1366.4	0.156	0.078801	(0.10)	0.62028	(0.30)	0.05709	(0.27)	**488.96**	**490.02**	**495.0**	0.43
z7	1.4	7.5	0.62	454.9	0.203	0.078501	(0.10)	0.62099	(0.59)	0.05737	(0.55)	487.17	490.46	505.9	0.44
z8	0.5	22.0	0.64	1289.9	0.203	0.078842	(0.09)	0.62054	(0.26)	0.05708	(0.23)	**489.21**	**490.19**	**494.8**	0.44
Gabbro, Marginal Way (MT01-34) (GPS coordinates: N42° 25.577′, W70° 55.105′)															
z1	1.3	75.9	0.87	4179.3	0.273	0.078713	(0.08)	0.61779	(0.16)	0.05692	(0.13)	**488.43**	**488.46**	**488.6**	0.55
z2	0.3	948.8	0.78	53186.7	0.245	0.078719	(0.09)	0.61783	(0.13)	0.05692	(0.09)	**488.47**	**488.49**	**488.6**	0.68
z3	0.6	288.9	0.87	15878.2	0.272	0.078735	(0.08)	0.61784	(0.13)	0.05691	(0.10)	**488.57**	**488.49**	**488.2**	0.65
z4	0.6	305.1	0.77	17133.4	0.243	0.078750	(0.09)	0.61852	(0.14)	0.05696	(0.11)	**488.66**	**488.92**	**490.1**	0.62
Syenodiorite, Little Nahant (MT06-02) (GPS coordinates: N42° 26.273′, W070° 55.988′															
z1	0.4	18.6	0.80	1055.1	0.254	0.078993	(0.10)	0.62299	(0.35)	0.05720	(0.32)	**490.11**	**491.72**	**499.2**	0.42
z2	0.3	17.7	0.88	983.6	0.278	0.078989	(0.10)	0.62201	(0.34)	0.05711	(0.31)	**490.08**	**491.11**	**495.9**	0.39
z3	0.3	13.0	0.76	747.5	0.240	0.079056	(0.12)	0.62232	(0.54)	0.05709	(0.50)	**490.49**	**491.30**	**495.1**	0.43
z4	0.3	32.1	0.85	1779.8	0.271	0.078799	(0.09)	0.62025	(0.27)	0.05709	(0.24)	488.95	490.00	494.9	0.46

Note: Corr. coef.—correlation coefficient. Age calculations are based on the decay constants of Steiger and Jäger (1977).
[†]All analyses are single zircon grains and pre-treated by the thermal annealing and acid leaching (CA-TIMS) technique. Data used in age calculations are in bold.
[§]Pb_c is total common Pb in analysis. Pb* is radiogenic Pb concentration.
[#]Measured ratio corrected for spike and fractionation only.
[††]Radiogenic Pb ratio.
[§§]Corrected for fractionation, spike, blank, and initial common Pb. Mass fractionation correction of 0.25%/amu ± 0.04%/atomic mass unit (amu) was applied to single-collector Daly analyses. Total procedural blank averaged 1.0 pg for Pb and less than 0.1 pg for U. Blank isotopic composition: $^{206}Pb/^{204}Pb$ = 18.27 ± 0.1, $^{207}Pb/^{204}Pb$ = 15.59 ± 0.1, $^{208}Pb/^{204}Pb$ = 38.12 ± 0.1.

$^{206}Pb/^{238}U$ date to be generally the most precise and accurate representation of age.

Detailed U-Pb date errors are reported as ±X/Y/Z in Table A1 (Appendix 1), where X is the internal (analytical) uncertainty in the absence of all external errors, Y incorporates the U-Pb tracer calibration error, and Z includes the latter as well as the decay constant errors of Jaffey et al. (1971). The external uncertainties must be taken into account, if the results are to be compared with dates obtained in other laboratories or dates derived from other isotopic chronometers (e.g., U-Pb versus ^{40}Ar-^{39}Ar). Therefore, only Z is reported in Figures 3A–3D and discussed in this section. However, where comparison is made among the results of this study alone (see section under Intrusive History), only internal errors need be considered. Full discussion of both analytical methods and error calculations can be found in Thompson et al. (2007).

TABLE A1. COMPLETE U-Pb DATE ERRORS FOR ALKALIC IGNEOUS ROCKS AT NAHANT

Sample no.	Rock type	Weighted mean $^{206}Pb/^{238}U$ date ± error*	MSWD[†]/(n)[§]
MT98-16	Syenite	488.48 ± 0.18/0.25/0.79	0.19 (5)
MT01-32	Felsic dike	489.03 ± 0.21/0.28/0.82	0.39 (5)
MT06-02	Syenodiorite	490.19 ± 0.29/0.36/0.90	0.63 (3)
MT01-34	Gabbro	488.53 ± 0.20/0.27/0.81	0.22 (4)

*All errors are 2σ.
[†]Mean square of weighted deviates.
[§]Number of analyses included in average.

Gabbro at Marginal Way (MT01-34)

Coarse-grained gabbro at this locality contains steeply north-dipping layers defined by concentrations of feldspar. Angular granitoid xenoliths measuring up to 40 cm are also present. The sampled gabbro is dominated by plagioclase (An_{61}) and diopsidic pyroxene, with minor olivine and interstitial biotite. Secondary minerals include serpentine rimming or replacing olivine, brown amphibole replacing pyroxene, muscovite ± calcite replacing plagioclase, and disseminated clots of chlorite and epidote. Apatite, ilmenite, magnetite, pyrite, titanite, and zircon are present as accessories. CA-TIMS analyses of four single zircons from this sample overlap within uncertainty and yield a weighted mean $^{206}Pb/^{238}U$ date of 488.53 ± 0.81 Ma (mean square of weighted deviates [MSWD] = 0.22) (Fig. 3A), which we interpret as the crystallization age.

Syenite at Saunders Ledge (MT98-16)

This sample comes from the northeastern coast of Nahant where pink syenite appears both to cut and be cut by gabbro and finer grained mafic rocks. The complex, mutually intrusive relationships within this striking assemblage indicate mingling between coexisting liquids in the same magma chamber. The syenite contains turbid, altered perthitic alkali-feldspar with interstitial quartz and albite locally verging on granophyric intergrowths. Clusters of ferroactinolite and ferroaugite along with accessory apatite, magnetite, titanite, and zircon occur throughout. The weighted mean $^{206}Pb/^{238}U$ date from five single zircon analyses is 488.48 ± 0.79 Ma (MSWD = 0.19) (Fig. 3B), which we interpret as the crystallization age of the mingled magmas at Saunders Ledge.

Syenodiorite at Little Nahant (MT06-02)

This dark gray, pink weathering syenodiorite forms a sill lying structurally below the Cambridge Argillite and mafic sills at Johns Peril (Fig. 1). The interior of the sill is coarser than the lower margin where the sample was collected for dating. This sample contains medium-grained plagioclase phenocrysts altered to mixtures of albite, muscovite, and alkali-feldspar in a

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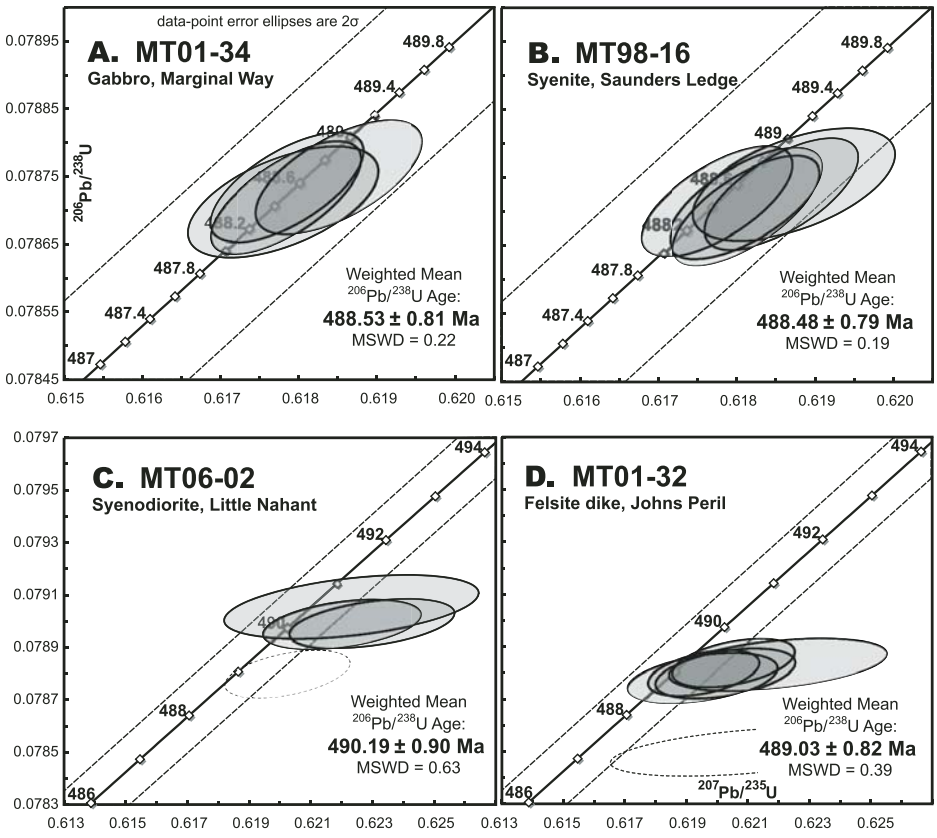


Figure 3. U-Pb concordia plots for Nahant samples from locations shown in Figure 1. Ages in Ma are marked on the concordia curve, and individual analyses of single zircons are shown as two-sigma error ellipses. Shaded ellipses signify analyses used in age calculation. Dotted lines mark the concordia error envelope. Complete U-Pb isotopic data are listed in Table 1.

matrix of albite, quartz and alkali-feldspar, and minor chloritized biotite. Clots of chlorite + titanite + apatite probably pseudomorph primary pyroxene (reported by Verma, 1973), and other secondary minerals include hematite, calcite, and barite (in veins). Accessory minerals include apatite, magnetite, and slender prismatic zircon. Four single zircons were analyzed from this sample, three of which overlapped within error and gave a weighted mean $^{206}Pb/^{238}U$ date of 490.19 ± 0.90 Ma (MSWD = 0.63) (Fig. 3C). The fourth analysis yielded a younger date interpreted to represent Pb loss and was excluded from age calculation.

Felsic Dike at Johns Peril (MT01-32)

The sea cliff at Johns Peril on the north side of Nahant (Fig. 1) displays Cambridge Argillite hornfels intruded by a SE-dipping gabbroic sill, as well as a 6-m-thick, ENE-trending felsic dike (keratophyre of Lane, 1890; granite in Fig. 2). All of these units are truncated by the gabbro of central Nahant. Quartz stringers in the hornfelsed argillite and a faint fabric in the dike both trend ENE, subparallel to the contact with the gabbro.

The dated sample from the felsite dike contains fine-grained intergrowths of quartz + alkali-feldspar + albite with shreds of chloritized biotite and a few plagioclase phenocrysts converted to phengitic K-mica + calcite. Accessories include apatite, magnetite, monazite, rutile, xenotime, and zircon. Five out of six analyzed single zircons from this sample formed a coherent cluster with a weighted mean $^{206}Pb/^{238}U$ date of 489.03 ± 0.82 (MSWD = 0.39), with one younger analysis interpreted to reflect Pb loss (Fig. 3D). We interpret 489.03 ± 0.82 Ma as the crystallization age of the dike, which also provides a maximum age for the crosscutting gabbro.

Intrusive History

Since all of the dates presented in this paper were produced in the same laboratory and following the same set of procedures, the external errors can be ignored and an intrusive history can be constructed using only internal errors (±X in Table A1). Calculated weighted mean $^{206}Pb/^{238}U$ dates from the four samples do not overlap within analytical error, and the sequence that emerges is both consistent with field observations of the Nahant intrusions and demonstrates that these intrusions were very closely spaced in time.

The syenodiorite sill at Little Nahant (490.19 ± 0.29 Ma) is the oldest intrusive phase at Nahant. Undated gabbroic sills within the Cambridge Argillite at Johns Peril are older than the 489.03 ± 0.21 Ma felsic dike that cuts them both. All units at Johns Peril, together with mafic sills penetrating the Lower Cambrian Weymouth Formation at East Point, are in turn cut by gabbro underlying most of Nahant and associated syenite at Saunders ledge (488.53 ± 0.20 Ma and 488.48 ± 0.18 Ma, respectively). The current outcrop area of gabbro and syenite, which coincides with a positive aeromagnetic anomaly centered on Nahant (Harwood and Zietz, 1976), is interpreted as representing only the steep-sided feeder plug to a formerly more extensive body of magma that ponded above the stratified country rocks and early intruded syenodiorite and gabbro sills, thus causing them to sag up to 45° (East Point in cross section of Fig. 1). Such tilting in conjunction with incremental magmatic injection has been suggested to account for the dip of early formed layers in other plutons on a variety of scales. Documented examples include the Jurassic Pyramid Peak granite and associated dioritic rocks that crop out over an area measuring ~2 km × 6 km in the northern Sierra Nevada, California (Wiebe et al., 2002), the 14 km × 27 km, Silurian Cadillac Mountain Intrusive Complex in the Coastal Maine Magmatic Province (Wiebe et al., 1997), and comparably sized Cretaceous plutons of the Coast Plutonic Complex in British Columbia (Brown and McClelland, 2000).

This synthesis expands upon an earlier interpretation of relationships at East Point that links Nahant magmatism with crustal downwarping (Ross and Bailey, 2001; Kelly and Ross, 2002).

PALEOMAGNETIC INVESTIGATION AT NAHANT

Paleomagnetic sampling for this study was conducted at 12 sites (BOS4A through BOS4L) mostly within gabbro and diabases intruding stratified sections at East Point–Great Ledge and Johns Peril where tilt corrections can be applied. These sites are shown by filled circles in Figure 1, and their geographic coordinates obtained from a handheld global positioning system are listed in Table 2.

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TABLE 2. PALEOMAGNETIC DATA FROM NAHANT, MASSACHUSETTS

Site number	Rock type	Latitude °N	Longitude °W	NRM $\times 10^{-2}$ A/m	χ $\times 10^{-5}$ SI	*n/N*	Bedding	Component	No tilt correction D	I	α_{95}	k	VGP PPlong	PPlat	Tilt-corrected D^t	I^t	α_{95}	k	VGP $PPlong^t$	$PPlat^t$	Plat
BOS4A*	Gabbro	N42° 24.999′	W070° 54.366′	319.8	2732	7/7	234/44W	A	319.8	–34.9	3.5	301	149	18	306.4	–78.4	3.5	300	309	27	68
BOS4B*	Gabbro	N42° 25.196′	W070° 54.279′	321.1	361	7/9	243/35W	A	321.1	–42.4	6.9	78	145	14	296.6	–75.2	6.8	78	317	26	62
BOS4D*	Gabbro	N42° 25.095′	W070° 54.368′	301.7	3804	7/7	228/43W	A	301.7	–37.9	5.3	130	162	7	259.4	–75.0	5.3	129	327	41	62
BOS4F*	Gabbro	N42° 25.785′	W070° 55.520′	251.9	676	6/6	023/15E	A	251.9	–89.0	3.5	358	292	43	290.6	–74.2	3.5	359	320	27	60
BOS4G*	Gabbro	N42° 25.785′	W070° 55.520′	228.9	4055	7\8	023/15E	A	228.9	–87.3	5.3	132	295	46	284.4	–73.6	5.3	133	323	29	60
BOS4H*	Gabbro	N42° 25.785′	W070° 55.520′	171.1	1036	8/8	023/15E	A	171.1	–84.4	4.2	178	286	53	271.1	–77.1	4.2	172	321	37	65
BOS4J*	Gabbro	N42° 25.298′	W070° 54.543′	192.8	2706	4/9	Variable	A	192.8	–58.2	43.4	6	35	80	247.5	–72.8	9.3	100	334	46	58
BOS4K	Gabbro			----	3389	9	----		----	----	----	----	----	----	----	----	----	----	----	-----	----
BOS4C	Mafic dike	N42° 25.194′	W070° 54.279′	162.7	3145	6/7	235/32W	C	162.7	9.1	12.9	28	132	40	167.9	39.2	12.9	28	121	24	22
BOS4E	Mafic dike	N42° 25.092′	W070° 54.537′	300.4	8376	3/4	228/43W	A	300.4	–43.3	8.6	204	161	3	239.6	–77.0	8.6	202	323	51	65
BOS4I	Felsic dike	N42° 25.785′	W070° 55.520′	334.0	21	11/11	023/15E	pdf (low T)	334.0	68.2	6.4	52	235	70	15.8	75.8	6.4	52	308	67	63
BOS4I	Felsic dike			176.1		10/11	023/15E	B (high T)	176.1	59.8	6.5	57	112	7	158.3	50.9	6.5	57	128	13	32
BOS4L	Syenite	N42° 25.498′	W070° 54.572′	337.9	6084	8/9	023/15E	pdf (low T)	337.9	77.1	4.6	144	268	64	55.1	79.3	4.6	143	317	51	69
BOS4L	Syenite			165.2		6/9	023/15E	B (high T)	165.2	51.0	9.2	54	122	15	153.9	40.6	9.2	54	135	20	23
	Mean gabbro (7 sites*)					**N = 7**			298.1	–70.1	25.9	6			**278.3**	**–76.0**	**4.2**	**208**	**321**	**34**	**64**
	Mean gabbro and BOS4E (8 sites)					**N = 8**			298.7	–66.4	22.9	7			**274.0**	**–76.4**	**4.2**	**174**	**321**	**36**	**64**
Data from Weisse et al. (1985)																					
Site 6	Syeno-diorite	N42° 26.102′	W070° 55.775′			6	023/15	A	224.4	82.2	9.7	48.6	278	30	144.2	75.9	9.7	48.6	306	19	63
Site 7	Gabbro	N42° 25.547′	W070° 55.595′			6		A	273.1	–74.8	4.5	217.9	144	–35	273.1	–74.8	4.5	218	325	35	62
	Mean Nahant (gabbro and Weisse sites)					N = 9			298.6	–73.6	19.5	8			**282.4**	**–76.3**	**3.9**	**174**	**320**	**32**	**64**
	Mean Nahant (gabbro, Weisse sites, and BOS4E)					N = 10			299.0	–70.3	18.3	8			**278.5**	**–76.7**	**3.9**	**152**	**320**	**34**	**65**

Mean poles	**VGPs**	**Long.**	**Lat.**	**A_{95}**
Mean gabbro (this study)	N = 7	321	34	7.7
Mean gabbro and BOS4E (this study)	N = 8	321	36	7.7
Mean Nahant (gabbro and Weisse sites)	N = 9	320	32	7.1
Mean Nahant (gabbro, Weisse sites, and BOS4E)	**N = 10**	**320**	**34**	**7.2**

Note: Site coordinates were determined from handheld global positioning system Magellan receiver of place name geographic directory; site locations are shown in Figure 1. NRM—natural remanent magnetization; χ—Initial susceptibility; *n/N*—number of samples used for site mean calculation/total number of samples; Bedding—strike/dip; D—site mean declination; I—site mean inclination; α_{95}—semi-angle of the cone of 95% confidence about the site mean; Plat—paleolatitude; VGP—virtual geomagnetic pole in Northern Hemisphere; PPlat—paleopole latitude; PPlong—paleopole longitude; A_{95}—radius of error circle at the 95% confidence level; positive values are in the Northern Hemisphere, and negative values are in the Southern Hemisphere. pdf—present-day field: 344.5°, 69°N.

*Directions used to calculate the Nahant mean.

[t]Tilt-corrected values.

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Seven or more cores were obtained with a Pomeroy gasoline-powered drill at most of the sampling sites where drilling was possible, and these were oriented with a Brunton magnetic compass on a Pomeroy orienting fixture. Oriented hand specimens were collected at sites that could not be drilled. Sun compass measurements were made at most localities to allow correction for local magnetic variation, but none was observed. Structural observations were used, where feasible, to conduct paleomagnetic tests assessing possible remagnetization events.

Specimens were cut from more than 100 cores, and natural remanent magnetization was measured using an AGICO JR5A magnetometer in the Byrd Polar Research paleomagnetism laboratory, Ohio State University. Pilot samples were demagnetized from each site using both alternating field and thermal techniques. The cleaning technique that yielded the most linear demagnetization trajectories was chosen for use on the remaining samples from each site. Alternating field demagnetization was carried out on a Schonstedt GSD-1 instrument in 2–3 mT steps up to ~15 mT, followed by steps of 5–10 mT up to a maximum of 95 mT (minimum of 10 steps). For thermal demagnetization, samples were heated in an ASC TD48 oven in 50 °C steps up to 500 °C, followed by 10–20 °C temperature steps (minimum of 14 steps). After each thermal demagnetization step, samples were evaluated for mineralogical changes using a Bartington MS2 susceptibility meter. Magnetization directions were determined by principal component analysis (Kirschvink, 1980) and considered to be stable (whether primary or secondary), if defined by three or more collinear points with a maximum angular deviation value of <15° (usually <10°) when plotted on vector component diagrams.

A previous paleomagnetic investigation at Nahant (Weisse et al., 1985) produced structurally uncorrected results from seven sites within the Nahant Gabbro and one site in the syenodiorite at Little Nahant (numbered open circles in Fig. 1). These will be discussed along with the results from the current study.

Primary Magnetization Direction

In all samples analyzed from Nahant, the remanent magnetization was removed below 590 °C or below 100 mT. These unblocking temperatures along with coercivities of the samples indicate that hematite was not a magnetic carrier. Typical demagnetization curves for samples from several sites in the gabbro appear in Figure 4. Generally, a consistent A component of magnetization was removed between 500 °C and 590 °C, indicating titano-magnetite as the main carrier of the magnetization. At temperatures below ~500 °C, gabbro samples produced linear trajectories that were not sufficiently consistent within site to allow a secondary component to be calculated. These lower temperature components are commonly viscous remanents

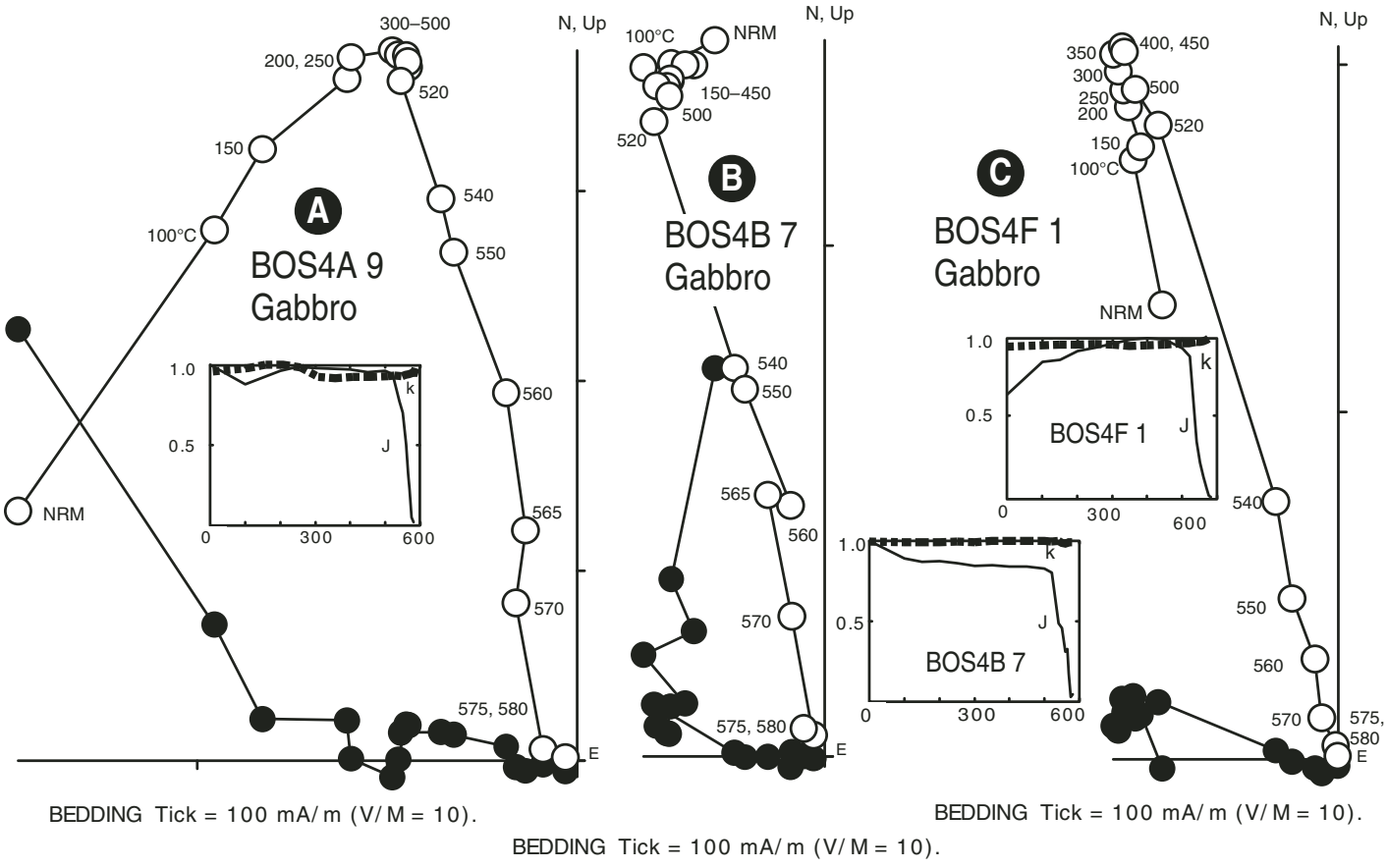


Figure 4. Orthogonal projection of vector endpoint diagrams showing demagnetization behavior of gabbro samples at Nahant. Open circles (solid circles) are projections on the vertical (horizontal) planes at the indicated levels of alternating field (AF) (mT, milliTesla) or thermal (°C) demagnetization. Normalized intensity (J/Jo) plots and susceptibility (k) versus temperature plots are shown for samples where thermal demagnetization was used. Diagrams A, B, and C show A components of magnetization at three numbered sites (locations in Fig. 1). NRM—natural remanent magnetization.

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of the present-day field or possibly overprints from younger tectonic events, as will be discussed for syenodiorite and diorite members of the Nahant suite. Results from site BOS4K did not yield stable, internally consistent within-site directions, hence are not considered in the analysis below.

Structurally uncorrected directions from gabbro sites listed in the top of Table 2 fall into two main groups (Fig. 5A). Northwest-trending and moderately upward-pointing magnetic directions were obtained from three dipping sills (sites BOS4A, BOS4B, and BOS4D) and a mafic dike (site BOS4E) sampled on the eastern end of Nahant. The second group of directions based on three dipping gabbroic sills at Johns Peril on the north coast of the island (sites BOS4F, BOS4G, and BOS4H) is south- to southwest-trending and very steeply upward-pointing. A third direction, south-trending and moderately steeply upward-pointing (Fig. 5A), was obtained from four of nine cores drilled at site BOS4J where sheeting joints in the main body of Nahant gabbro have been deformed into south-verging kink folds. All of these directions converge into a single group once the sills and dike at East Point–Great Ledge and Johns Peril are untilted (based on attitudes included in Table 2) and sheeting joints at BOS4J are unfolded. With these structural corrections, the mean direction based on eight sites is D = 274.0°, I = –76.4° (α_{95} = 4.2°, N = 8) and the associated paleopole is 321°E, 36°N (Table 2).

Paleomagnetic directions have previously been reported by Weisse et al. (1985) from five locations in the Nahant Gabbro. Three of these are based on data combined from two drilling sites (1+2, 3+5, and 8+9 in Fig. 1). These results show poor within-site consistency and are not further considered. Results from single locations in both the gabbro and the syenodiorite (sites 7 and 6 in Fig. 1), however, define tight groupings (respective α_{95} values of 9.7° and 4.5°) that can be compared and integrated with site means obtained in our study. Site 7 is located in the parking lot behind the Nahant Police Department, within the feeder plug of Nahant Gabbro that remains in its original orientation so that no structural correction is necessary. The mean direction from this site (D and I in Table 2 from Table 1 in Weisse et al., 1985) plots together with our structurally corrected directions from other localities (Fig. 5B). The mean direction from the syenodiorite at Little Nahant (D and I in Table 2 from Table 1 in Weisse et al., 1985) has been tilt corrected in accordance with our interpretation of this body as a sill that sagged into its present orientation beneath later and higher injection(s) of gabbro. The structurally corrected site mean from this locality is D = 144.2°, I = 75.9°, essentially antipodal with respect to mean directions in the gabbro (Fig. 5B).

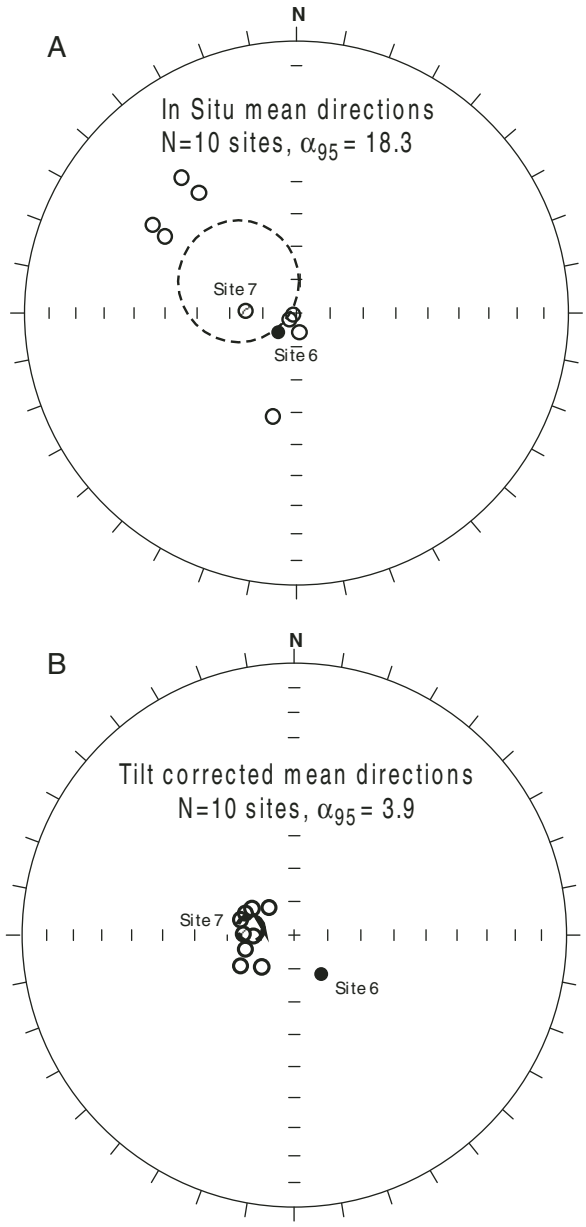


Figure 5. Equal angle projections of site mean directions based on A component magnetizations at Nahant (Table 2). Open (closed) symbols are on the upper (lower) hemisphere. Circle represents the α_{95} radius of 95% confidence for the directional mean. With tilt correction for all sites except Site 7 (Weisse et al., 1985), mean direction is D = 278.5°, I = –76.7° (α_{95} = 3.9°, N = 10).

Combining appropriately corrected results from these two sites of Weisse et al. (1985) with our structurally corrected results from eight sites gives rise to a mean direction of D = 278.5°, I = –76.7° (α_{95} = 3.9°, N = 10). These results yield a statistically significant passage of the fold test of McElhinny (1964) and McFadden and Jones (1981) because they are best grouped after 100% unfolding. A reversals test on the Weisse et al. (1985) sites 6 and 7 yields a critical angle of 12.6°, constituting a positive reversals test with an Rc classification (McFadden and McElhinny, 1990). Given the positive fold test and reversals test, we consider the corresponding paleopole at 320°E, 34°N (A_{95} = 7.2°, N = 10) to represent a primary magnetization direction in the Nahant intrusive suite (Table 2).

The 490–488 Ma crystallization ages from the Nahant suite span the most recent estimate for the Cambro-Ordovician boundary (488.3 ± 1.7 Ma in Gradstein et al., 2004). The new paleopole places Nahant at a paleolatitude of approximately 65°S ± 7° at this time.

Secondary Magnetization Directions

Nongabbroic rocks at Nahant contain magnetization directions distinct from those in the gabbros (Fig. 6). A low-temperature component in both the 489 Ma felsite dike at Johns Peril (site

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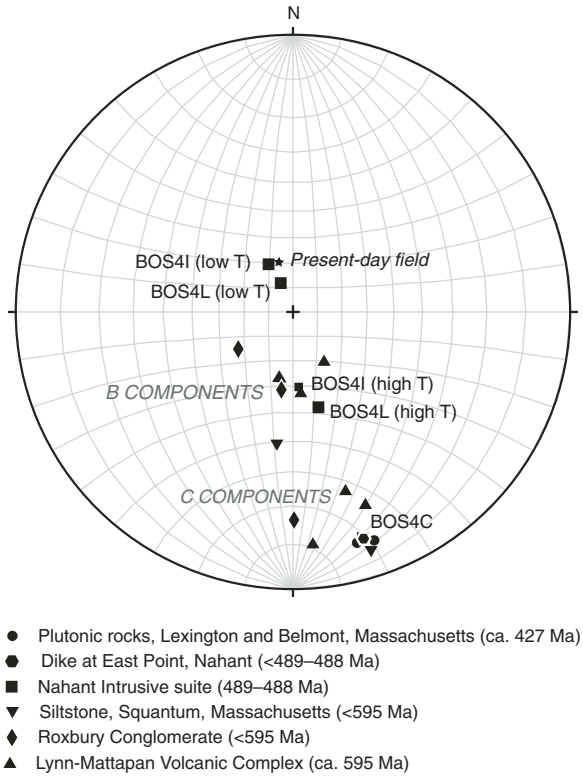


Figure 6. Magnetization directions at sites BOS4C, BOS4I, and BOS4L represent the present-day field as well as secondary B and C components found in Ediacaran to Silurian-age rocks in the greater Boston area. Individual site directions are plotted for the Lynn-Mattapan Volcanic Complex and Squantum siltstone (Table 2 in Thompson et al., 2007); the Roxbury Conglomerate (sites 4, 6, and 9 of Fang et al., 1986); and plutonic rocks around Belmont-Lexington, Massachusetts (Hurley and Shearer, 1981).

BOS4I in Fig. 1 and Table 2) and the 488 Ma syenite at Saunders Ledge (site BOS4L) represents viscous overprinting from the Earth's present-day magnetic field (D = 344°E, I = 69°N). High-temperature components at both of these sites are south-southeast to south-trending and moderately downward pointing and plot with "B components" recognized in the 595 Ma Lynn-Mattapan Volcanic Complex (Thompson et al., 2007). The 100 million year age range of rocks containing this magnetization direction indicates that it is of secondary origin at Nahant.

Similar arguments apply to a south-southeast and gently downward-pointing direction in a mafic dike at East Point, Nahant (site BOS 4C in Fig. 1 and Table 2). This direction resembles the "C component" in both the Lynn-Mattapan Volcanic Complex and younger siltstone at Squantum Head, Massachusetts (Thompson et al., 2007; Fig. 6; maximum age 595 Ma in Thompson and Bowring, 2000). A similar direction is reported by Hurley and Shearer (1981) in mafic plutonic rocks west of Boston now dated as Early Silurian (ca. 427 Ma U-Pb date of Hepburn et al., 1998). The dike at Nahant contains only this direction, suggesting that dike injection and associated fluid migration associated with this activity led to chemical remagnetization in the older rocks.

Two magnetic components reported in Roxbury Conglomerate and associated volcanic rocks, respectively, west and south of Boston, Massachusetts (Fang et al., 1986) also resemble B and C directions described above. Two of the three site means from this study for which α_{95} <16° plot with B components in Figure 6 after plunge and tilt corrections (recalculated using our own measurements at Fang et al.'s sites 6 and 9 because this information is not included in their Table 1). The third site yields an in situ site mean that is C-like in its gently downward pointing inclination (Fig. 6; site 4 in Fang et al., 1986). The age of the Roxbury Conglomerate remains problematic and could be significantly younger than the 595 Ma maximum age supplied by its Squantum "Tillite" Member (detailed discussion in Thompson and Grunow, 2004).

TECTONIC IMPLICATIONS

Gabbroic and syenodioritic to syenitic intrusions into strata including the Lower Cambrian shallow marine Weymouth Formation at Nahant, Massachusetts, record an episode of 490–488 Ma bimodal magmatism after Avalonian arc activity in southeastern New England had given way to deposition on a stable platform. The 490–488 Ma magmatic activity is consistent with diverse Cambro-Ordovician events in East Avalonia, Cadomia, Iberia, Bohemia, and Ganderia associated with breakup of the northern Gondwanan margin and separation of these peri-Gondwanan terranes from their original positions bordering northwestern Africa and Amazonia to form the Rheic Ocean. Late Cambrian to early Tremadocian sequences in the Welsh Basin and English Midlands of East Avalonia, for example, are viewed in terms of rapid subsidence governed by transtensional normal faulting early in the rifting process (Prigmore et al., 1997). Volcanic records heralding the tectonic transition include bimodal Middle to Upper Cambrian magmatic rocks in the Saxo-Thuringian Zone of the Bohemian Massif (Kemnitz et al., 2002) as well as a bimodal Middle Cambrian volcanic sequence in the Ossa-Morena Zone of southern Spain (Etxebarria et al., 2006). Late Cambrian plutonism in the Hermitage Flexure in Newfoundland (499 +3/–2 Ma granodiorite of Dunning and O'Brien, 1989; 499 ± 2 Ma gabbro of O'Brien et al., 1996) and in the Bras d'Or terrane of Nova Scotia (493 ± 2 Ma granite of Dunning et al., 1990) has been linked with the beginning of rifting of Ganderia from the Gondwanan margin (van Staal et al., 1998). The Cambro-Ordovician paleopole from Nahant confirms the peri-Gondwanan position of West Avalonia at the onset of this important new tectonic regime.

Rheic Paleogeography

In general agreement with earlier reconstructions based on paleomagnetic data combined with faunal arguments (Van der Voo, 1988; Scotese and McKerrow, 1990; Cocks and Fortey, 1990; van Staal et al., 1998; Cocks and Torsvik, 2002), our reconstruction using the Nahant paleopole as well as Early Ordovician reference poles from Gondwana (Grunow and Encarnacion, 2000) and North America (Mac Niocaill et al., 1997) shows Avalonia at a high southerly latitude bordering northern Gondwana, which itself borders on the south geographic pole (Fig. 7). Our results clarify and amplify the existing paleogeographic picture in several respects.

The broad acceptance that the opening of the Rheic Ocean was associated with the Lower Ordovician departure of Avalonia from northern Gondwana rests on surprisingly few unambiguous paleopoles that establish the starting position of Avalonia. Results from the Tremadoc–early Arenig Treffgarne Volcanic Formation, SW Wales (Trench et al., 1992) have been used in a 480 Ma reconstruction by Cocks and Torsvik (2002) to place southern Britain at paleolatitude of 62°S ± 9° in a position bordering NW Africa. To achieve the expected facing of the north English margin of Avalonia toward Iapetus, however, the original Treffgarne pole (56°N, 306°E, dp/dm = 9/10) was rotated through 180°

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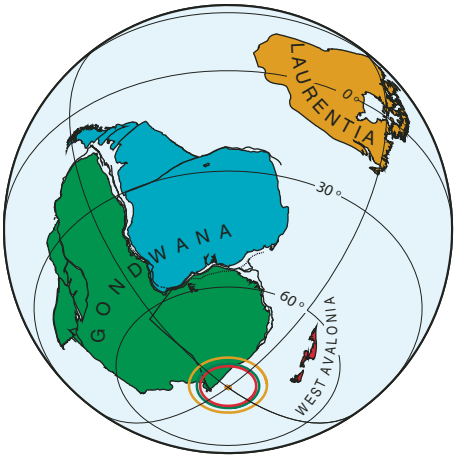


Figure 7. Plate reconstruction of West Avalonia at 490–488 Ma based on the Nahant paleopole at 320°E, 34°N (A_{95} = 7.2°, N = 10) and Early Ordovician Gondwanan reference poles from Grunow and Encarnacion (2000). North America reference poles from Mac Niocaill et al. (1997). A_{95} circles for West Avalonian, Gondwanan, and Laurentian paleopoles in red, green, and gold, respectively. The reconstruction is based on superimposing the centers of the A_{95} circles of West Avalonia, Gondwana, and North America.

(Fig. 12 in Trench et al., 1992). This correction was justified in terms of the ±22° declination error resulting from very steep magnetic inclinations in the Treffgarne samples and possible tectonic rotations affecting the survey area (Trench et al., 1992). These data also fix only the eastern end of the once presumably contiguous Avalonian belt that now continues as West Avalonian terranes from the Avalon Peninsula of Newfoundland to southeastern New England. The New England segment of West Avalonia can now be positioned using our new Nahant paleopole at a comparable paleolatitude of 65°S ± 7°, without need for a rotation correction.

From a starting position at approximately 65°S at ca. 490 Ma, West Avalonia can also be tracked as it separated from Gondwana during Rheic opening. West Avalonia had moved ~20°–25° northward by the Middle Ordovician based on paleomagnetic results from the Dunn Point Formation in the Avalon terrane of Nova Scotia. Basalts and rhyolites in this sequence yield a site mean direction with D = 344°, I = –60° (N = 19, α_{95} = 4.1°), corresponding to a paleopole at 130°E, 2°N and a paleolatitude of 41°S (Johnson and Van der Voo, 1990; latest Ordovician–earliest Silurian Rb-Sr date cited in the paleomagnetic study superseded by 460.0 ± 3.4 Ma U-Pb date of Hamilton and Murphy, 2004). The latitudinal component of drift between 490 Ma and 460 Ma averages 9.2 cm/a, a fast rate that seems reasonable in view of the 12.5 cm/a rate inferred for the closing of the Iapetan segment between the Notre Dame Arc and the Popelogan-Victoria Arc (equivalent to Exploits Arc of Mac Niocaill et al., 1997) that occurred between 470 and 455 Ma on the north side of Avalonia as the Rheic Ocean widened (Fig. 7d in van Staal et al., 1998). During the subsequent closure of the Tetagouche-Exploits backarc basin (including the Tetagouche Group investigated paleomagnetically by Liss et al., 1993) and Ganderia's approach to the Laurentian margin (Fig. 7e in van Staal et al., 1998; Hibbard et al., 2006; van Staal et al., 2008), Avalonia continued to converge with Laurentia, but more slowly. This rate has been estimated at 5.5 cm/a (Hamilton and Murphy, 2004) based on the 460 Ma Dunn Point volcanics (~41°S at 460 Ma) and the Saint Mary's gabbro sills in the Newfoundland Avalon zone (~32°S at 441 Ma from Hodych and Buchan, 1998; U-Pb baddeleyite date on Saint Mary's sills from Greenough et al., 1993). The collision between Avalonia and Laurentia began at ca. 421 Ma (van Staal et al., 2008), consistent with paleomagnetic arguments below.

Post-Cambrian Tectonic History

Magnetic B and C components from 490 to 488 Ma rocks at Nahant, Massachusetts (Table 2), and other Boston-area rocks of various age (Fig. 6) give rise to virtual geomagnetic poles (VGPs) that appear to reflect younger Paleozoic tectonic episodes in the Southeastern New England Avalon Zone. In Figure 8, these VGPs track time segments of the North American apparent polar wander path (after Van der Voo, 1990) that coincide with middle and late Paleozoic tectonism elsewhere in the northern Appalachians.

VGPs from structurally uncorrected directions in the Saunders Ledge syenite and the granitic dike at Johns Peril (sites BOS4I and BOS4L in Table 2) and from the Lynn-Mattapan Volcanic Complex (Thompson et al., 2007) fall along the Late Silurian to Lower Devonian segment corresponding to the interval when Avalonia is generally thought to have docked with Laurentia (Robinson et al., 1998; Hibbard, 2004; van Staal et al., 2008). The VGP calculated from the structurally corrected Roxbury Conglomerate direction at site 6 of Fang et al. (1986) also lies on this portion of the apparent polar wander path, while the structurally corrected VGP from site 9 is slightly removed from it (Fig. 8). One interpretation consistent with the Roxbury VGPs is that this prefolding magnetization direction represents remagnetization of Neoproterozoic rocks before the Southeastern New England Avalon Zone was part of North America (Fang et al., 1986). Should the Roxbury prove to be Paleozoic in age, the prefolding magnetization could also be a primary direction.

VGPs based on structurally uncorrected directions from a mafic dike at East Point (site BOS4C in Table 2), from the Lynn-Mattapan Volcanic Complex and Squantum siltstone (Thompson et al., 2007), from plutonic rocks around Belmont and Lexington, Massachusetts (Hurley and Shearer, 1981), and from site 4 in Roxbury-related volcanic rocks (Fang et al., 1986), all plot at higher latitudes (Fig. 8). Most of these cluster tightly along the Carboniferous segment of the North American apparent polar wander path (Fig. 8). This interval corresponds to the time of "Neo-Acadian" deformation and plutonism ascribed in Nova Scotia to oblique collision between the Meguma terrane and Avalonia (White et al., 2007). Widespread 366 Ma to 350 Ma plutonism, high-grade metamorphism, and ductile shearing in central New England demonstrate the broad reach of Neo-Acadian tectonism, although the tectonic setting remains poorly understood (Robinson et al., 1998). Avalonia and Ganderia were probably joined at this time, however, because Middle Cambrian siltstones from the Bourinot Group in Cape Breton Island, Nova Scotia (Bras d'Or terrane of White et al., 1994) contain an intermediate temperature, C-like magnetization direction for which the paleopole (126°E, 37°N of Johnson and Van der Voo, 1985 also plotted in Fig. 8) closely resembles the VGPs calculated from overprints throughout the greater Boston area.

CONCLUSION

Comagmatic gabbro, syenodiorite, and syenite at Nahant, Massachusetts, intrude fossiliferous Lower Cambrian platform deposits that mark the end of the Avalonian tectonic cycle in southeastern New England. Samples spanning this compositional range yield tightly clustered U-Pb single-zircon dates between 490 Ma and 488 Ma obtained using CA-TIMS methods. Ten paleomagnetic sampling sites from the gabbro and related sills show a regionally consistent A component of magnetization with a mean direction of D = 278.5°, I = –76.7° (α_{95} = 3.9°, N = 10) that passes both fold and reversals tests. This direction is interpreted to represent a primary magnetic direction acquired close to the crystallization ages within the intrusive suite. The mean paleopole of 320°, 34°N (A_{95} = 7.2°, N = 10) places the southeastern New England portion of West Avalonia at a paleolatitude of 65° at the time of the Cambro-Ordovician boundary and demonstrates a paleogeographic position

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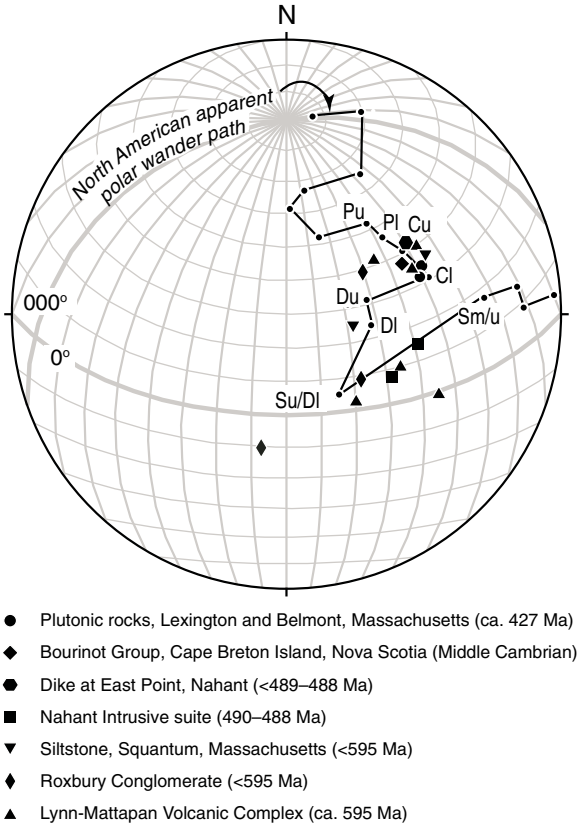


Figure 8. Magnetic B and C components from Nahant (this study), the Lynn-Mattapan Volcanic Complex and Squantum siltstone (Thompson et al., 2007), plutonic rocks around Lexington-Belmont, Massachusetts (Hurley and Shearer, 1981), and the Roxbury Conglomerate (Fang et al., 1986) give rise to VGPs that lie along the North American apparent polar wander path of Van der Voo (1990) after mid-Silurian time. The Bourinot Group from Cape Breton Island, Nova Scotia, is represented in the plot by the paleopole of Johnson and Van der Voo (1985). Sm/u—Middle to Upper Silurian; Su/Dl—Upper Silurian to Lower Devonian; Dl—Lower Devonian; Du—Upper Devonian; Cl—Lower Carboniferous; Cu—Upper Carboniferous; Pl—Lower Permian; Pu—Upper Permian.

at high latitude consistent with previous faunal arguments. This position slightly removed from the African margin of Gondwana, together with the bimodal character of the Nahant igneous assemblage, suggests that the Cambro-Ordovician magmatism is related to breakup of the Gondwanan margin and the opening of the Rheic Ocean. Virtual geomagnetic poles calculated from stable B and C magnetic components at Nahant sampling sites in syenodiorite and syenite lie along Late Silurian to Carboniferous segments of the North American apparent polar wander path. These and other VGPs in the Boston area indicate that West Avalonia had docked with Laurentia by that time.

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